Deep Learning Assignment Outline

1. Package Loading

- Load only the necessary packages, including TensorFlow, NumPy, and visualization tools like Matplotlib.

- Ensure compatibility with the hardware for optimal performance.

2. Data Loading

- Load the CIFAR-10 dataset, split into training and test sets.

- Display the initial dimensions of the dataset to confirm loading.

3. Exploratory Data Analysis (EDA)

3.1 Sample Visualization

- Display a 5x5 grid of sample images from the dataset, showcasing various classes.

3.2 Class Distribution

- Plot the class distribution across the entire dataset (both training and test).

3.3 Pixel Intensity Distribution

- Analyze and visualize the distribution of pixel intensities across each color channel (Red, Green, and Blue) to understand data characteristics.

4. Data Processing

4.1 Normalization

- Scale pixel values to the range [0, 1] to improve model convergence.

4.2 Splitting

- Split the test set into validation and test subsets (e.g., 50% for validation and 50% for test).

4.3 Confirm Data Dimensions

- Display dimensions of x_train, y_train, x_val, y_val, x_test, and y_test after processing to ensure everything is set up for training.

5. Model Training and Hyperparameter Optimization

- Train and optimize each model one by one, using validation data to assess performance.

5.1 ResNet-18 (from scratch)

- Define and train ResNet-18 architecture.
- Perform hyperparameter tuning (learning rate, batch size, epochs).
- Record training/validation accuracy and loss curves.

5.2 AlexNet (pretrained)

- Load and train a pretrained AlexNet model, fine-tuning it for CIFAR-10.
- Conduct hyperparameter optimization.
- Record training/validation accuracy and loss curves.

5.3 SimpleCNN (pretrained)

- Use a smaller pretrained CNN model suited for CIFAR-10.
- Perform hyperparameter tuning to assess improvements.
- Record training/validation accuracy and loss curves.

5.4 Two Additional Lightweight Networks

- Select two more small-scale networks (e.g., MobileNet, SqueezeNet).
- Apply similar training and optimization steps.
- Record metrics and curves for comparison.

6. Model Comparison and Final Testing

6.1 Validation Set Comparison

- Compare all models based on validation accuracy, loss, and efficiency.

- Choose the best-performing model based on these metrics.

6.2 Test Set Evaluation

- Use the selected model to evaluate performance on the test set.

- Report final accuracy, loss, and additional metrics (precision, recall, F1-score).

- Generate a confusion matrix for deeper insight into class-level performance.

7. Summary Overview

7.1 Summary of Results

- Summarize the results of each model, including accuracy, computational efficiency, and hyperparameter settings.

7.2 Best Model Discussion

- Provide insights into why the best model performed well, and discuss potential areas for improvement.

7.3 Future Directions

- Suggest possible future work, such as trying other architectures or tuning additional hyperparameters.